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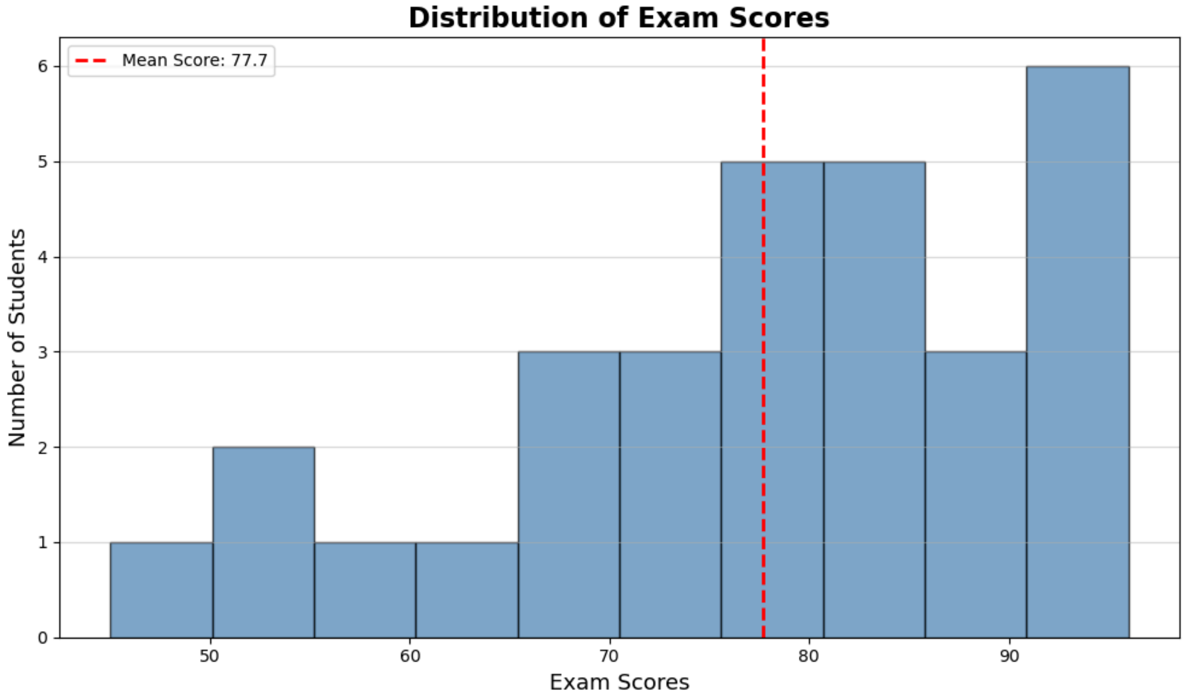
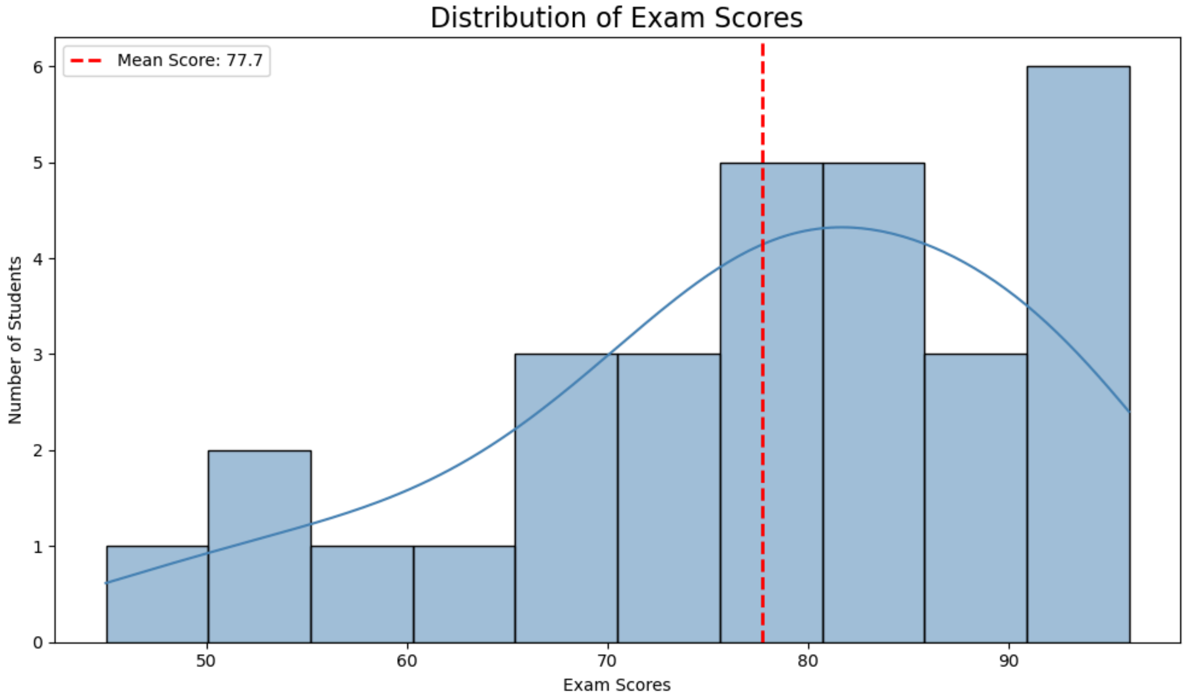
Question => Utilize Python data visualization libraries to create a histogram ilustrating the distribution of exam scores

Solution

In [1]:
```
import matplotlib.pyplot as plt
import numpy as np

exam_scores = [45, 52, 60, 63, 67, 70, 72, 74, 75, 76,
               78, 79, 80, 81, 82, 83, 85, 86, 88, 90,
               91, 92, 93, 94, 95, 96, 55, 68, 77, 84]

plt.figure(figsize=(10, 6))

plt.hist(exam_scores, bins=10, color='steelblue',
         edgecolor='black', alpha=0.7)

plt.title('Distribution of Exam Scores', fontsize=16, fontweight='bold')
plt.xlabel('Exam Scores', fontsize=13)
plt.ylabel('Number of Students', fontsize=13)

mean_score = np.mean(exam_scores)
plt.axvline(mean_score, color='red', linestyle='dashed',
            linewidth=2, label=f'Mean Score: {mean_score:.1f}')

plt.legend()
plt.grid(axis='y', alpha=0.5)
plt.tight_layout()
plt.show()
```



In [2]:
```
import seaborn as sns
import matplotlib.pyplot as plt

exam_scores = [45, 52, 60, 63, 67, 70, 72, 74, 75, 76,
               78, 79, 80, 81, 82, 83, 85, 86, 88, 90,
               91, 92, 93, 94, 95, 96, 55, 68, 77, 84]

plt.figure(figsize=(10, 6))
sns.histplot(exam_scores, bins=10, kde=True,
             color='steelblue', edgecolor='black')

plt.title('Distribution of Exam Scores', fontsize=16)
plt.xlabel('Exam Scores')
plt.ylabel('Number of Students')

mean_val = np.mean(exam_scores)
plt.axvline(mean_val, color='red', linestyle='dashed',
            linewidth=2, label=f'Mean Score: {mean_val:.1f}')
plt.legend()

plt.tight_layout()
plt.show()
```



CODE SUMMARY

1. Imported matplotlib and seaborn for visualization
2. Created sample exam scores dataset of 30 students
3. Plotted histogram using matplotlib with 10 bins
4. Plotted histogram using seaborn with KDE curve
5. Added Mean Score (77.7) as a red dashed reference

In [ ]: